Heston model: calibration to market data and exotic option pricing

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1 Introduction and goals

The preceding Heston notes take the model from theoretical foundations to four independent pricing methods: closed-form Fourier, full-truncation Euler MC, Andersen QE MC, and 2D PDE with Douglas ADI. On the benchmark strip of European calls ($K = 80, 85, \ldots, 120$, $T = 1$;

`python/results/phase4/heston_cross_validation.csv`) the PDE price is within 0.3% of the Fourier price (largest relative difference 2.9×10^{-3}, at $K = 120$) and both Monte Carlo estimates lie within their 95% confidence intervals of it. This note closes the cycle by addressing the two questions that practitioners actually face:

1. **Calibration.** Given a market-observed surface of European option prices (or equivalently, implied volatilities) at various strikes and maturities, find the Heston parameters $(\kappa, \theta, \sigma, \rho, v_0)$ that best reproduce this surface. The model must be *fitted to the market* before it can be used for anything else.

2. **Exotic option pricing.** Once the model is calibrated, price options whose payoffs are not European calls. Path-dependent payoffs (Asian, lookback, barrier) and early-exercise payoffs (American puts) are the typical workhorses of this category, and they cannot be priced by the closed-form Fourier method.

These two tasks form the core operational workflow of derivatives trading under Heston: *calibrate to liquid vanilla quotes, then price illiquid exotics with the calibrated model.* The accuracy of the entire workflow depends on both halves: a poorly-calibrated model gives wrong exotic prices, and a poorly-implemented exotic pricer fails to deliver the calibrated model's accuracy.

This note has four goals.

Calibration mathematics. Define the calibration problem as a constrained least-squares optimisation in parameter space. Identify the natural objective function (price RMSE or implied-vol RMSE), the parameter constraints (Feller condition, $\rho \in [-1, 1]$), and the practical optimisation strategies (Levenberg-Marquardt, differential evolution, hybrid).

Calibration robustness. The Heston calibration is notoriously ill-conditioned: multiple parameter sets reproduce a given vanilla surface to within market precision, and the optimiser may converge to local minima or to physically-unreasonable parameters. We discuss diagnostics and the use of regularisation.

Exotic pricing. Implement the arithmetic-average Asian call, the floating-strike lookback call, and the up-and-out barrier call using QE Monte Carlo (the natural workhorse for path-dependent European-style payoffs). For American puts, extend the 2D PDE pricer with a projection onto the early-exercise constraint after each time step (the 1D treatment by PSOR is in the note *Pricing the American put via PSOR*).

Validation. The validation strategy now must operate without the Fourier ground truth that anchored the European pricers. Cross-method consistency (PDE vs MC) becomes the primary diagnostic, complemented by analytical limit cases (e.g., a barrier with $H \to \infty$ recovers the European price).

2 The calibration problem

2.1 Formal statement

Given N market observations of European call prices $\{C_i^{\text{mkt}}\}_{i=1}^N$ at strike-maturity pairs $\{(K_i, T_i)\}_{i=1}^N$, find Heston parameters $\boldsymbol{\theta} = (\kappa, \theta, \sigma, \rho, v_0) \in \Theta$ that minimise an aggregate distance between model

and market:

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}\in\Theta} \; \mathcal{L}(\boldsymbol{\theta}), \tag{1}$$

where $\mathcal{L}$ is an objective function (loss) and Θ is the admissible parameter set.

2.2 Choice of objective function

The two natural choices are price RMSE and implied-volatility RMSE:

$$\mathcal{L}_{\text{price}}(\boldsymbol{\theta}) = \sum_{i=1}^{N} w_i \left(C_i^{\text{model}}(\boldsymbol{\theta}) - C_i^{\text{mkt}}\right)^2, \tag{2}$$

$$\mathcal{L}_{\text{IV}}(\boldsymbol{\theta}) = \sum_{i=1}^{N} w_i \left(\sigma_i^{\text{model}}(\boldsymbol{\theta}) - \sigma_i^{\text{mkt}}\right)^2, \tag{3}$$

where $\sigma_i = \text{BSImpVol}(C_i, K_i, T_i, S_0, r)$ is the Black-Scholes implied volatility extracted from a price.

Price loss (2) is computationally cheap: each evaluation of C_i^{model} is one call to the Fourier pricer, no IV inversion needed. The gradient with respect to parameters is straightforward via finite differences or implicit differentiation. The drawback is that prices have wildly different magnitudes across the surface (deep ITM options have prices $\sim 10\times$ larger than ATM, and far-OTM options have prices that can be 0.1% of ATM), so the optimiser puts disproportionate weight on the most expensive options unless the weights w_i are calibrated to flatten this. Since $\Delta C_i \approx \text{vega}_i \, \Delta\sigma_i$, the choice $w_i = 1/\text{vega}_i^2$ approximately equalises the contribution of each observation per unit change in IV.

IV loss (3) is more expensive: each evaluation requires inverting the BS formula numerically (typically by Newton-Raphson, 5-10 iterations). But it is naturally on the right scale — IVs are all in the same order of magnitude — and is the metric traders actually care about, since vol surfaces are quoted in IV space. For our implementation we use the price loss with inverse-vega weights, $w_i = \text{vega}_i^{-1} / \max_j \text{vega}_j^{-1}$, where vega_i is the Black-Scholes vega at the market implied volatility (floored at 10^{-3}); the residuals passed to the optimiser are $\sqrt{w_i}\,(C_i^{\text{model}} - C_i^{\text{mkt}})$. Up to a constant factor the objective is therefore $\sum_i (C_i^{\text{model}} - C_i^{\text{mkt}})^2 / \text{vega}_i \approx \sum_i \text{vega}_i \, (\sigma_i^{\text{model}} - \sigma_i^{\text{mkt}})^2$: a compromise between the price and IV losses, computationally tractable, which still gives more weight to high-vega quotes than the IV loss would.

2.3 Parameter constraints

The physical constraints on the Heston parameters are $\kappa, \theta, \sigma > 0$, $v_0 \geq 0$ and $\rho \in [-1, 1]$. The admissible set Θ used by the Python implementation is the box

$$\begin{aligned}
\kappa &\in [0.01, 20] \text{ (mean-reversion speed)}, \\
\theta &\in [0.001, 1] \text{ (long-run variance)}, \\
\sigma &\in [0.01, 3] \text{ (vol of vol)}, \\
\rho &\in [-0.999, 0.999], \\
v_0 &\in [10^{-4}, 1] \text{ (initial variance)},
\end{aligned}$$

(`DEFAULT_BOUNDS` in `quantlib.heston_calibration`, overridable through the `bounds` argument), which keeps the optimiser away from the degenerate edges. The C++ implementation imposes only positivity and $|\rho| < 1$, through a change of variables (§2.4).

The Feller condition $2\kappa\theta \geq \sigma^2$ is sometimes imposed as an additional constraint for theoretical reasons (it ensures the variance process never reaches zero almost surely), but is widely violated by real-market calibrated parameters: empirically $2\kappa\theta < \sigma^2$ is the rule rather than the exception in equity markets. We do *not* enforce Feller. The calibration result does not report the Feller ratio $2\kappa\theta/\sigma^2$; it follows directly from the returned parameters.

2.4 Optimisation strategy

The Heston calibration objective (2) is generally non-convex with multiple local minima. Three layers of optimisation strategy address this:

Levenberg-Marquardt (LM). Local nonlinear least-squares optimiser; very fast but converges to the nearest local minimum. Excellent for refinement once we are close to the true minimum. The Python implementation calls `scipy.optimize.least_squares` with `method='trf'` (trust-region reflective), a local trust-region least-squares method that, unlike scipy's `'lm'`, accepts the bounds of §2.3. The C++ implementation is a hand-written LM with Marquardt's damping $\lambda\,\mathrm{diag}(J^\top J)$, a central-difference Jacobian and a 5×5 Cholesky solve of the damped normal equations; it works in the unconstrained coordinates $(\log\kappa, \log\theta, \log\sigma, \operatorname{artanh}\rho, \log v_0)$ instead of under bounds.

Differential evolution (DE). Global stochastic optimiser; slower but escapes local minima. Use as a coarse search to find the basin of the global minimum, then refine with LM. We use `scipy.optimize.differential_evolution`.

Hybrid (DE + LM). The standard recipe in production: run DE for a coarse global search (a few hundred function evaluations), then hand off the result to LM for high-precision local refinement. This combines the global reach of DE with the speed of LM.

For the implementation in this note we provide only the local optimiser (TRF in Python, LM in C++), which works well for synthetic test problems where the initial guess is reasonable; the `calibrate_heston` docstring sketches how to wrap it in DE for production use.

2.5 Identifiability concerns

The Heston model has 5 parameters, but they are not equally identifiable from a finite vanilla surface. Common patterns:

- v_0 is well-determined by short-maturity ATM options ($v_0 \approx \sigma^2_{\mathrm{ATM}} \cdot (1 + \mathcal{O}(T))$).
- θ is well-determined by long-maturity ATM options.
- ρ controls the slope of the volatility skew; well-determined when the surface includes wing strikes (deep OTM puts/calls).
- σ controls the convexity of the vol smile; harder to identify if the surface is sparse in strikes.
- κ is the most difficult to identify. It appears in combination with θ (in $\kappa\theta$) and with σ (in σ^2/κ as the long-run variance of variance). Different (κ, θ, σ) triples can produce nearly indistinguishable vanilla prices.

The practical consequence: do not over-fit. A calibrated $\kappa = 10$ when the surface contains only ATM points is an artifact, not a structural fact about the market. Parameter standard errors (from the inverse Hessian of the objective at the minimum) reported along with the point estimate would make this visible; the implementation does not compute them, and the parameter-stability check of §5.1 (Test 3 of `validate_heston_calibration.py`) is the diagnostic used instead.

3 Implied volatility inversion

We need a fast and reliable inversion of the Black-Scholes formula for the implied vol associated with a given call price.

3.1 The Newton-Raphson scheme

For a given $(K, T, S, r, C^{\text{mkt}})$, find σ such that $C^{\text{BS}}(K, T, S, r, \sigma) = C^{\text{mkt}}$. Newton-Raphson iteration:

$$\sigma_{n+1} = \sigma_n - \frac{C^{\text{BS}}(\sigma_n) - C^{\text{mkt}}}{\text{vega}(\sigma_n)}, \tag{4}$$

where vega $= \partial C^{\text{BS}}/\partial\sigma = S\sqrt{T}\,\phi(d_1)$ with ϕ the standard normal density and d_1 as in the BS formula.

From $\sigma_0 = 0.2$ the iteration is fast (typically fewer than ten iterations to a price residual below 10^{-8}), but it is not globally convergent: from $\sigma_0 = 0.01$ or $\sigma_0 = 5$ the vega is small compared with the price error, and the unguarded iteration breaks down within one or two steps (an overshoot leaves $(0, \infty)$, or the vega underflows to zero). The implementation therefore starts from $\sigma_0 = 0.2$ by default, clamps every Newton iterate to $[0.001, 5]$, and hands over to the bracketed fallback of §3.2 when the vega falls below 10^{-12} or Newton has not converged within 50 iterations. For the initial guess, the Manaster-Koehler formula provides a good starting point, but for our use case a constant $\sigma_0 = 0.2$ works fine.

3.2 Edge cases

Two pathologies worth noting:

- Prices below the intrinsic lower bound $\max(S - Ke^{-rT}, 0)$ have no admissible IV; the inversion must signal an error.
- Prices above the trivial upper bound S also have no admissible IV.
- For deep OTM options where the price is very close to zero, vega is also very small, leading to numerical instability of the Newton iteration. Mitigation: add a fallback to Brent's method for safety.

For the calibration loop, we use Newton with a fallback: if Newton fails to converge, switch to Brent's bracketed method on $[10^{-4}, 5.0]$ (bisection on the same bracket in C++), which is slower but converges whenever the implied volatility lies in the bracket; otherwise NaN is returned.

4 Exotic options under Heston

Once the model is calibrated, the next task is to price options whose payoffs are not European. We focus on three categories: path-dependent European-style (Asian, lookback), barrier (knock-out/knock-in), and early-exercise (American).

4.1 Asian options (arithmetic average)

The arithmetic-average Asian call has payoff

$$\Phi_{\text{Asian}}(\{S_t\}_{t\in[0,T]}) = \max\left(\frac{1}{n_{\text{avg}}}\sum_{k=1}^{n_{\text{avg}}} S_{t_k} - K,\ 0\right), \tag{5}$$

where $\{t_k\}$ are the averaging dates (typically uniformly spaced, e.g., daily over the option's life). The payoff depends on the entire path, so Fourier pricing does not apply; we use Monte Carlo.

The natural method is QE (see the QE note): the path-outer iteration of the simulator naturally accumulates the path average as it goes. The terminal-only QE simulator of `quantlib.heston_qe` is replaced by a slight modification that, instead of returning only S_T, accumulates $\bar{S} = \frac{1}{n_{\text{avg}}}\sum_k S_{t_k}$ during the iteration and returns $\bar{S}$.

4.2 Lookback options (floating-strike)

The floating-strike lookback call has payoff

$$\Phi_{\text{LBfloat}}(\{S_t\}_{t\in[0,T]}) = S_T - \min_{0\le t\le T} S_t. \tag{6}$$

Like the Asian, this is path-dependent, and Monte Carlo via QE is the natural method. The simulator accumulates $S_{\min}$ during iteration and returns it together with S_T.

The continuous-monitoring bias. The discrete-time simulation only observes the path at the time-step granularity $\Delta t = T/n_{\text{steps}}$. The minimum over a discretely-sampled path is biased upwards relative to the true continuous-time minimum (the path can dip lower between samples). For Brownian-like processes the bias is $O(\sqrt{\Delta t})$, which can be substantial. Production pricers use the continuity correction of Broadie, Glasserman and Kou [2] (introduced for discretely monitored barriers and extended by the same authors to lookbacks), which we mention as a future refinement.

4.3 Barrier options

The up-and-out call has payoff

$$\Phi_{\text{UO}}(\{S_t\}_{t\in[0,T]}) = \max(S_T - K, 0)\cdot \mathbb{1}\left\{\max_{0\le t\le T} S_t < H\right\}, \tag{7}$$

where $H > S_0$ is the barrier. The option pays the European call payoff only if the path never crosses the barrier; otherwise it "knocks out" to zero.

Like the lookback, the maximum is biased downward by discrete monitoring; the Brownian-bridge correction (Beaglehole-Dybvig-Zhou or similar) reduces this bias. The Monte Carlo simulation tracks the running max during iteration and applies the indicator at the end.

4.4 American options

American puts cannot be priced by Monte Carlo on a forward-only path, since the optimal exercise decision at each time depends on the conditional distribution of future paths. Two methods:

- **Longstaff-Schwartz Monte Carlo (LSM)**: simulates paths and uses regression on the path state to approximate the continuation value, then exercises optimally backward in time. Standard in MC-only environments.

- **2D PDE with projection**: extends the Douglas ADI pricer of the Heston PDE note with the early-exercise constraint. A rigorous treatment solves a linear complementarity problem inside each implicit sweep (PSOR, as in the 1D PSOR note); the simpler operator-splitting alternative takes one European Douglas step and then projects onto the constraint, $W \leftarrow \max(W, e^{r\tau}\Phi_{\text{exercise}})$.

We use the PDE with operator-splitting projection because (a) the Douglas ADI pricer of the Heston PDE note is already available, (b) the projection adds a single line to its time loop, and (c) PDE methods give Greeks for free. The price is a splitting error that vanishes as $\Delta\tau \to 0$; the early-exercise premium increases with N_τ refinement, and PSOR inside each sweep would be the more accurate choice. The implementation is `quantlib.heston_american_pde`.

5 Validation strategy

The validation is harder than for the European pricers because most exotics have no Fourier ground truth.

5.1 Calibration validation

Round-trip test. Generate synthetic "market" prices from a known set of Heston parameters, then run the calibration on those prices and verify that the recovered parameters match the originals to within numerical precision. This is the cleanest test of the calibration machinery.

Residual analysis. On a real (or simulated-as-real) surface, plot the per-strike, per-maturity residuals after calibration. Systematic patterns (e.g., residuals that increase with T) indicate either (a) the Heston model genuinely cannot fit the surface beyond the discrepancy floor, or (b) the optimiser converged to a local minimum.

Parameter stability. Run the calibration on slightly perturbed market data and verify that parameters change continuously with the data. Sudden jumps indicate the optimiser is bouncing between local minima.

5.2 Exotic pricing validation

Limit cases.

- Asian with $n_{\text{avg}} = 1$ at $t_1 = T$: same as European call.
- Up-and-out call with $H \to \infty$: same as European call.
- Down-and-out call with $H \to 0$: same as European call.
- American put with high mean-reversion ($\kappa \to \infty$ at fixed σ): the variance is pinned at θ, so the price tends to the Black-Scholes American put with volatility $\sqrt{\theta}$, not to the European put; the early-exercise premium does not vanish in this limit. The anchor here is a 1D American pricer (the PSOR note), not Fourier. The validation suite does not run this limit; it checks the deep out-of-the-money put instead (below).

The first three limits give us anchored validation against Fourier (`quantlib.heston_fourier`) for the corresponding European prices.

Cross-method consistency.

- For the American put, which has no Fourier reference: American $\geq$ European, intrinsic value deep in the money, closeness to the European price deep out of the money, and convergence under grid halving.
- Not implemented: for barrier options where there exists a closed-form analogue under BS (using the reflection principle), one could compare BS-with-Heston-IV vs Heston-via-MC. The discrepancy would be informative about the impact of stochastic vol on barrier pricing.

Convergence in n_{paths}. Standard MC convergence ($1/\sqrt{M}$) for QE-based exotics. Plot half-width vs $M^{-1/2}$ and verify linearity.

6 Implementation outline

6.1 Python

The module `quantlib.heston_calibration` exposes:

- `implied_vol_bs(C, K, T, S0, r)`: BS implied vol via Newton-Raphson with Brent fallback.
- `calibrate_heston(market_data, S0, r, initial_guess=None, *, bounds=None, weighted=True, max_iter=200, verbose=False)`: full calibration pipeline (inverse-vega weights when `weighted`, TRF under the bounds of §2.3). Returns a `CalibrationResult` named tuple with fields `params`, `residuals`, `rmse`, `n_iter`, `success`, `message`.

The module `quantlib.heston_exotics` exposes:

- `mc_asian_call_heston(S0, K, v0, r, kappa, theta, sigma, rho, T, n_steps, n_paths, *, n_avg=None, ...)`: arithmetic-average Asian call.
- `mc_lookback_call_heston(S0, v0, r, kappa, theta, sigma, rho, T, n_steps, n_paths, *, ...)`: floating-strike lookback.
- `mc_barrier_call_heston(S0, K, H, v0, r, kappa, theta, sigma, rho, T, n_steps, n_paths, *, ...)`: up-and-out barrier call.
- `heston_american_put_pde(S0, K, T, kappa, theta, sigma, rho, v0, r, ...)`, in the separate module `quantlib.heston_american_pde`: American put via the 2D PDE with operator-splitting projection. The module repeats the Douglas ADI kernels of `quantlib.heston_pde` so that it is self-contained.

The MC exotics share a common pattern: a path-aware simulator that accumulates the path-functional during the time loop. The QE step of `quantlib.heston_qe` is copied into `quantlib.heston_exotics` as a private kernel that the three simulators call, so that the module is self-contained.

6.2 C++

Same structure, with everything in the namespace `quant::heston`: `heston_calibration.hpp` declares `implied_vol_bs` and `calibrate_heston` (the hand-written LM of §2.4), and `heston_exotics.hpp` the three Monte Carlo pricers, which take the Heston parameters as a `HestonParams` struct and an explicit `std::mt19937_64` generator. The American pricer (`heston_american_pde.hpp`) repeats the Douglas ADI step of `heston_pde.cpp` and adds the projection inside the time loop.

6.3 Tests

The Catch2 tests cover part of §5. `test_heston_calibration.cpp`: the round trip on a 9-quote synthetic surface (parameters to 10^{-3} or 10^{-4}, RMSE below 10^{-4}), the success flag, implied-vol recovery to 10^{-7} on six cases, NaN outside the no-arbitrage bounds, and input validation. `test_heston_exotics.cpp`: the Asian ($n_{\text{avg}} = 1$) and barrier ($H = 10^6$) limits against the Fourier call within three half-widths, Asian and barrier ($H = 130$) prices below the European call by more than three half-widths, $1/\sqrt{M}$ scaling of the half-width for all three pricers, the positive discrete-monitoring bias of the lookback, and input validation. `test_heston_american_pde.cpp`: American $\geq$ European at $K = 80, 100, 120$, intrinsic value deep in the money ($K = 120$), early-exercise premium below 5% of the European price deep out of the money ($K = 70$), error reduction by more than a factor 2 under grid halving, finite prices within the loose bounds $(-0.01, K + 0.5]$, and input validation. There are no cross-method (MC against PDE) or large-κ tests.

7 Bibliographical notes

- Cui, del Baño Rollin, Germano (2017) [4]: Heston calibration with explicit gradients, the modern reference.
- Cont, da Fonseca (2002) [3]: empirical analysis of vol surface dynamics, motivates calibration methodology.
- Glasserman (2004) [5]: standard reference for MC pricing of path-dependent options including Asian, lookback, and barrier.
- Longstaff, Schwartz (2001) [6]: the LSM method for American options.
- Achdou, Pironneau (2005) [1]: PDE methods including PSOR for early exercise.

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